SDN MINI PROJECT REPORT:=

Project Title:
Topology Change Detector using SDN (Mininet + Ryu)

Subject Code: CN Mini Project

Name: SAIFULLA
SRN: PES2UG25AM807
Section: A

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1. Problem Statement

In traditional networks, detecting topology changes such as link failures is slow and complex.

The goal of this project is to design an **SDN-based system** that:

- Detects network topology changes dynamically
- Updates network state in real time
- Maintains efficient communication between hosts

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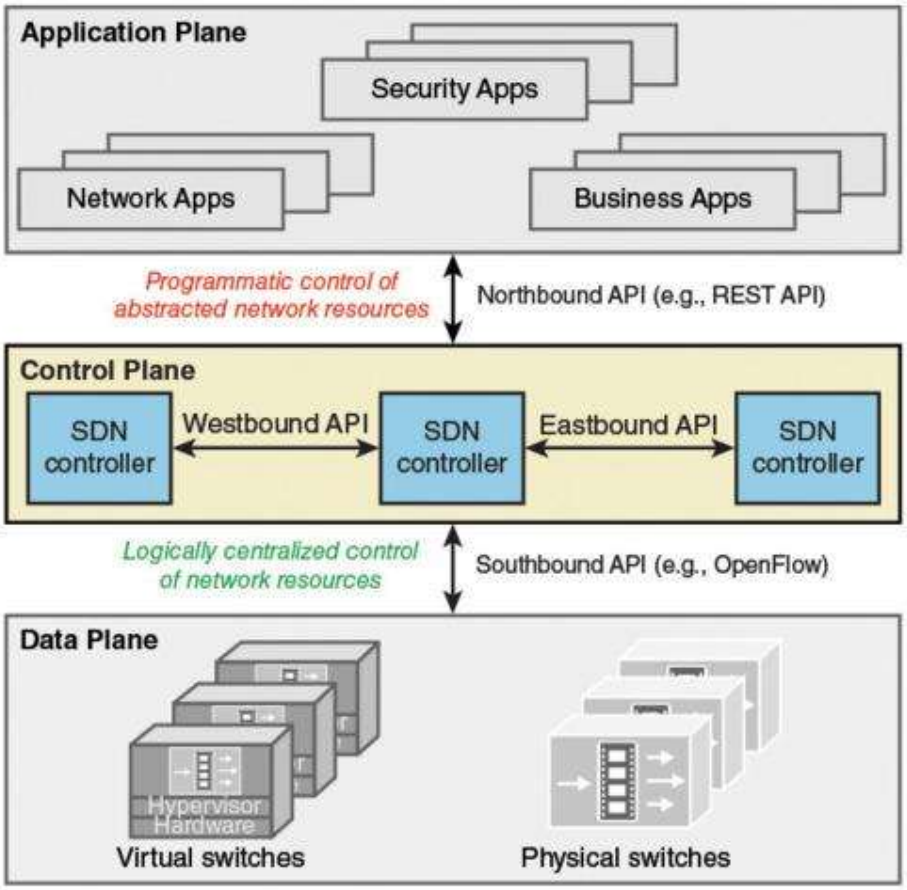
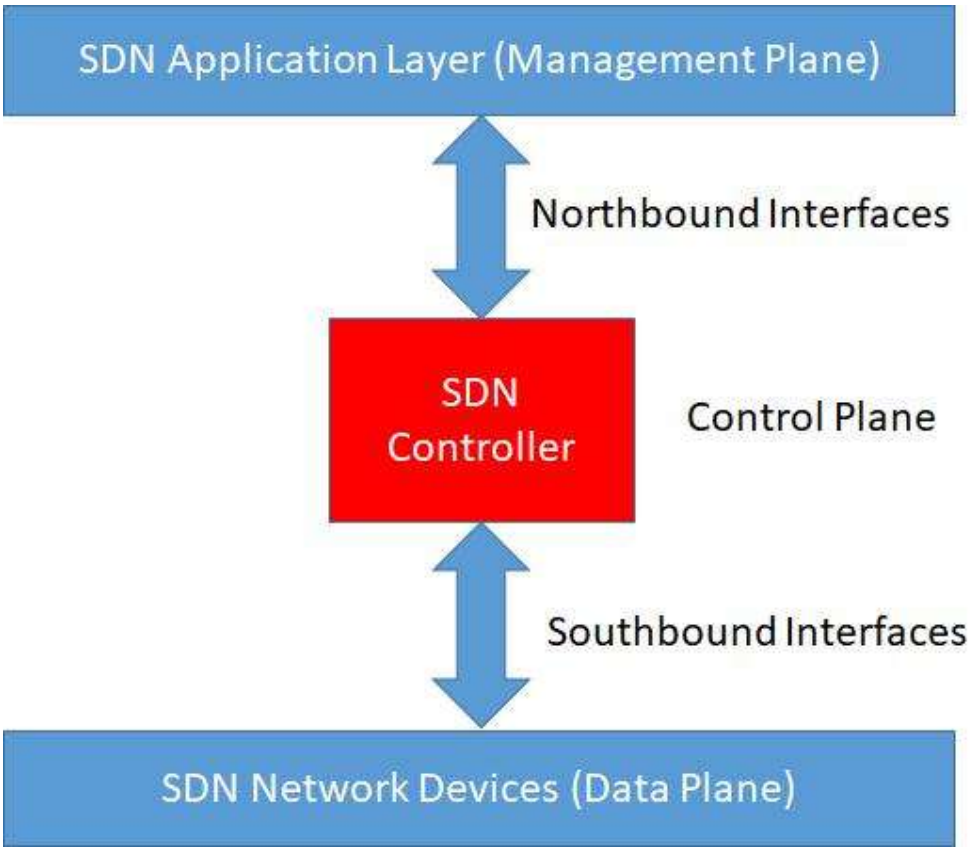
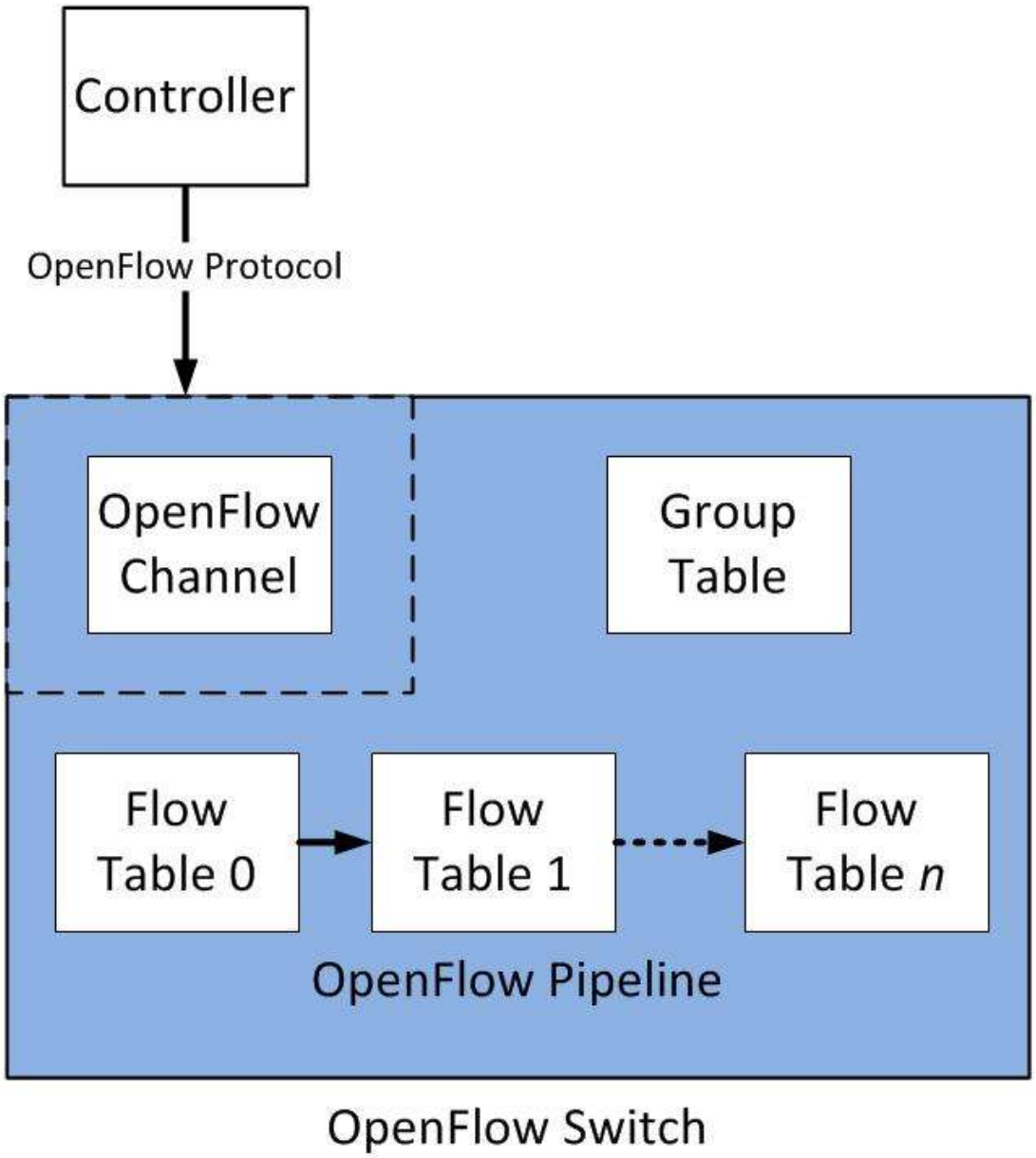
2. SDN Architecture




SDN Application Layer (Management Plane)
Northbound Interfaces
SDN Controller
Control Plane
Southbound Interfaces
SDN Network Devices (Data Plane)



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Software Defined Networking (SDN) separates:

- **Control Plane** → Ryu Controller
- **Data Plane** → Open vSwitch (Mininet switches)

Communication happens using **OpenFlow 1.3 protocol**.

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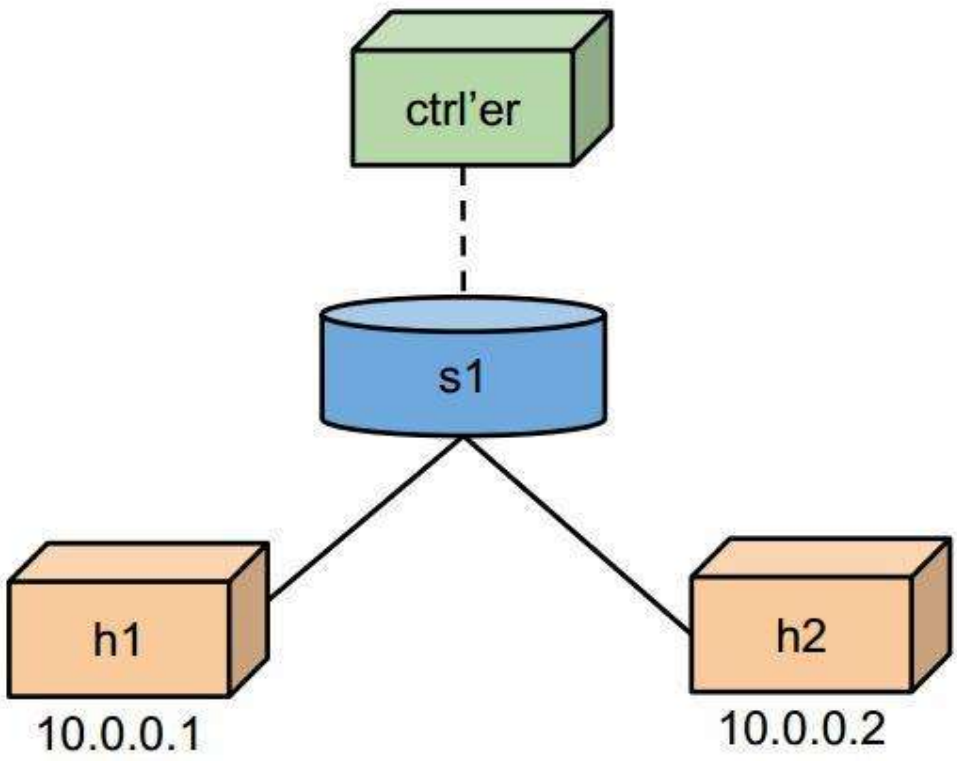
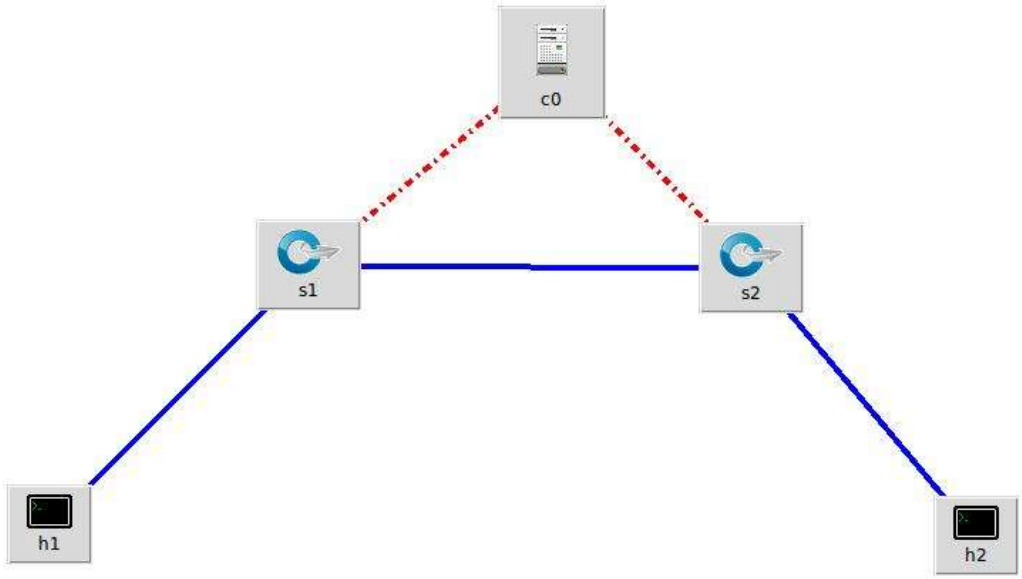
3. Mininet Topology Design





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Topology used:

- 2 Switches → s1, s2
- 2 Hosts → h1, h2

- Linear connection:
  h1 — s1 — s2 — h2

Command used:

```
sudo mn --topo linear,2 \
--controller=remote,ip=127.0.0.1,port=6653 \
--switch ovsk,protocols=OpenFlow13
```

---

4. Controller Implementation

The controller is implemented using **Ryu**.

Features:

- Detect switch connection
- Monitor links dynamically
- Handle packet_in events
- Install flow rules

Run command:

```
python3 -m ryu.cmd.manager --observe-links topo_detect.py
```

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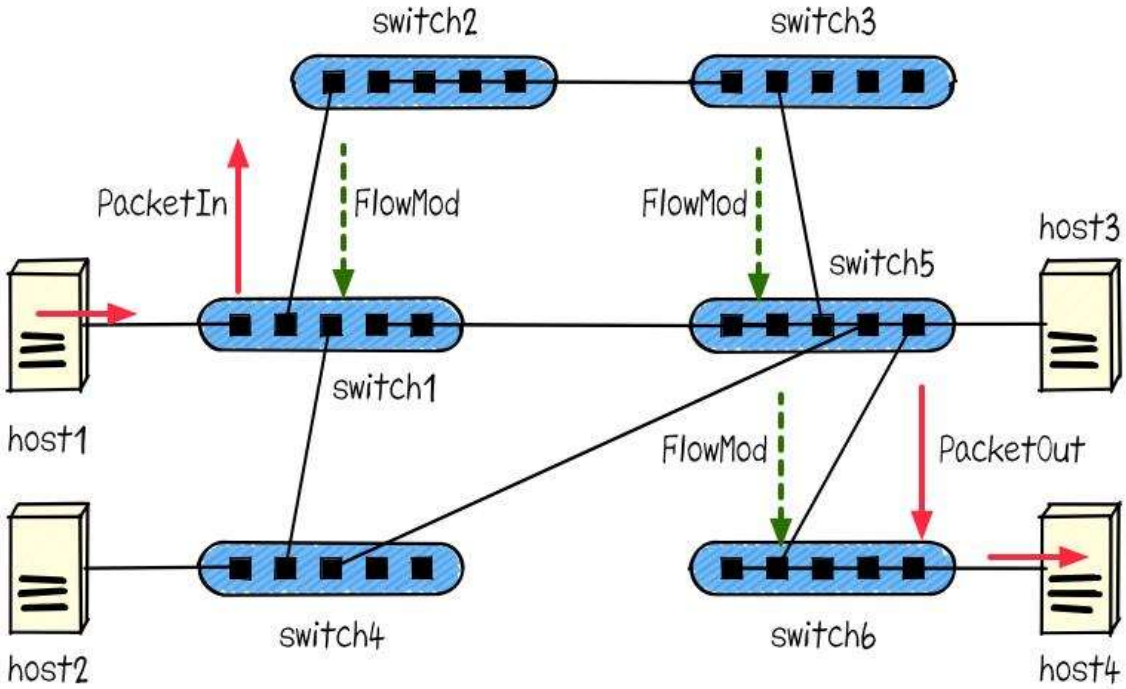
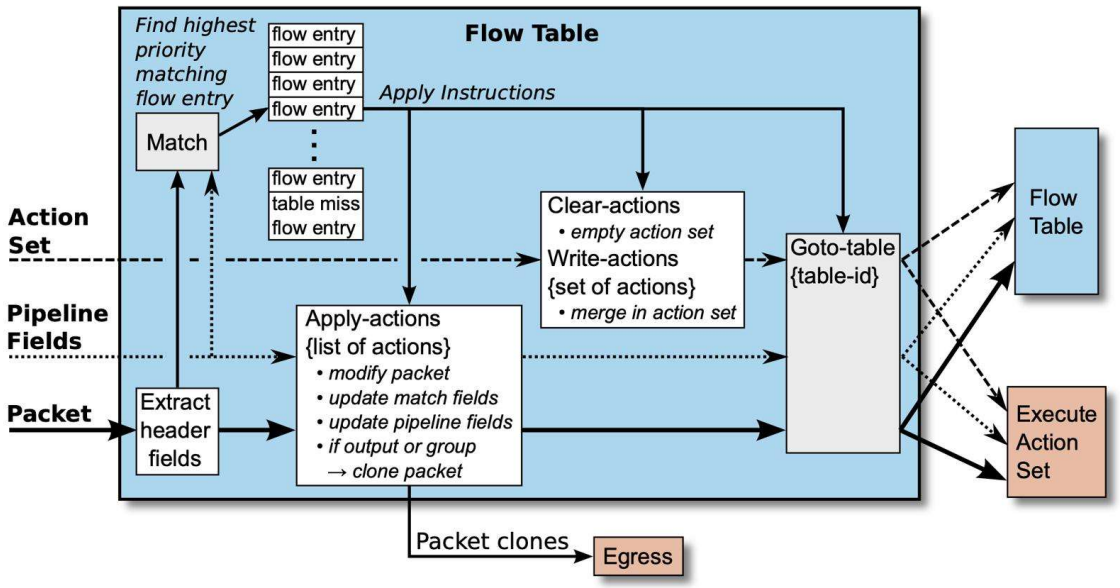
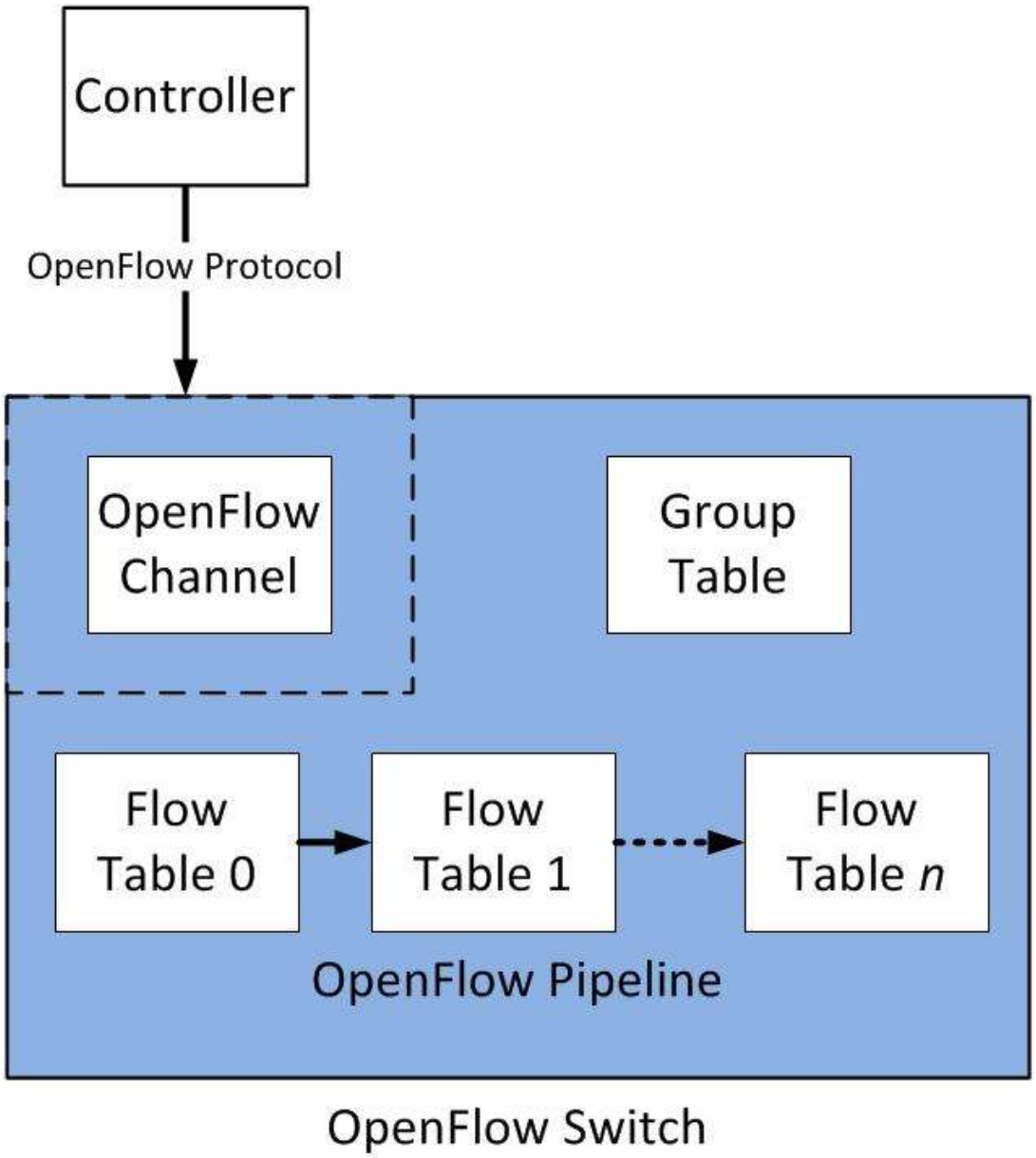
5. Flow Rule Management

Flow rules are installed using **OpenFlow (OFPFlowMod)**.


switch2
switch3
PacketIn
FlowMod
FlowMod
host3
switch5
switch1
host1
FlowMod
PacketOut
switch4
switch6
host2
host4


Find highest priority matching flow entry
Flow Table
flow entry
flow entry
flow entry
flow entry
Apply Instructions
Match
flow entry
table miss flow entry
Action Set
Clear-actions
• empty action set
Write-actions
{set of actions}
• merge in action set
Goto-table
{table-id}
Flow Table
Pipeline Fields
Apply-actions
{list of actions}
• modify packet
• update match fields
• update pipeline fields
• if output or group
→ clone packet
Packet
Extract header fields
Execute Action Set
Packet clones
Egress



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Working:

- First packet → sent to controller (packet_in)
- Controller learns MAC address
- Installs rule (FLOW_MOD)
- Next packets → directly forwarded

---

6. Functionality Implementation

✓ Learning Phase

Command:

mininet> pingall

- Unknown packets → controller
- MAC learning happens
- Flow rules installed

✓ Normal Operation

- All hosts communicate successfully
- No controller involvement after learning

✓ Topology Change Detection

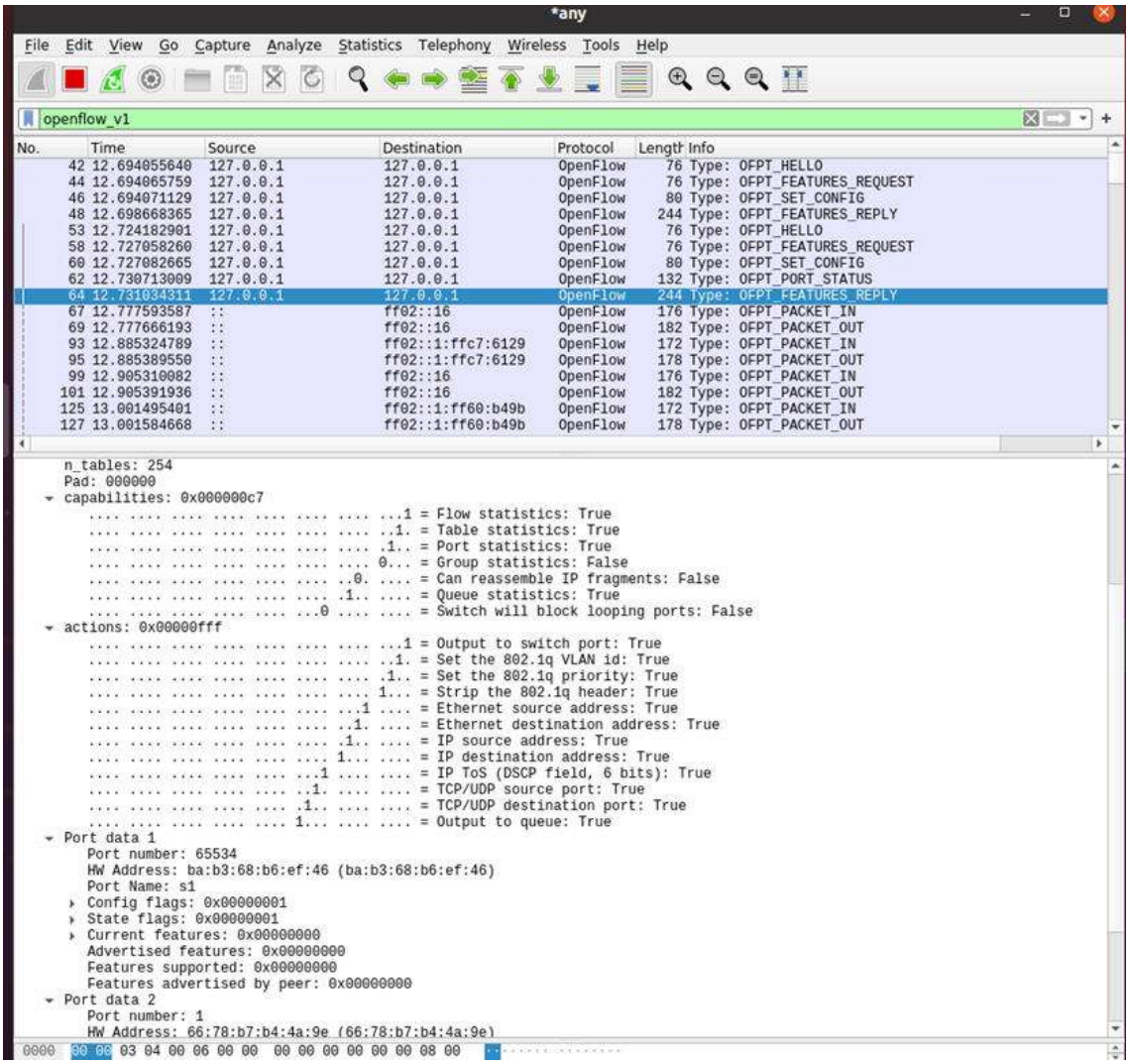
Command:

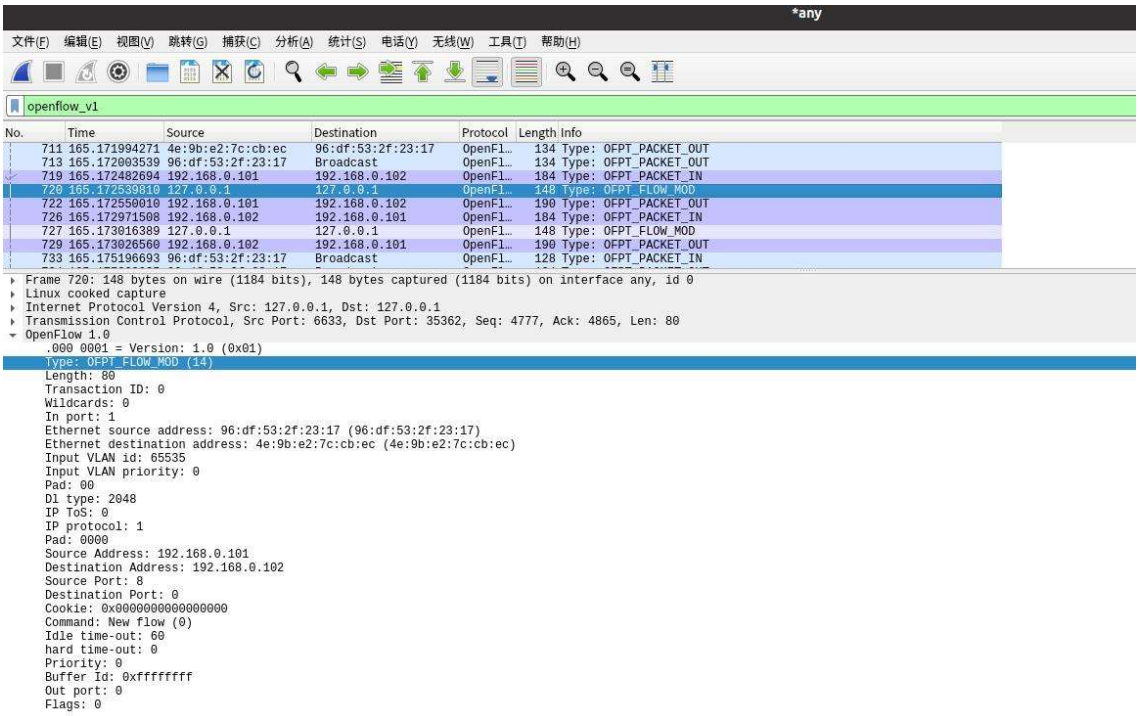
mininet> link s1 s2 down

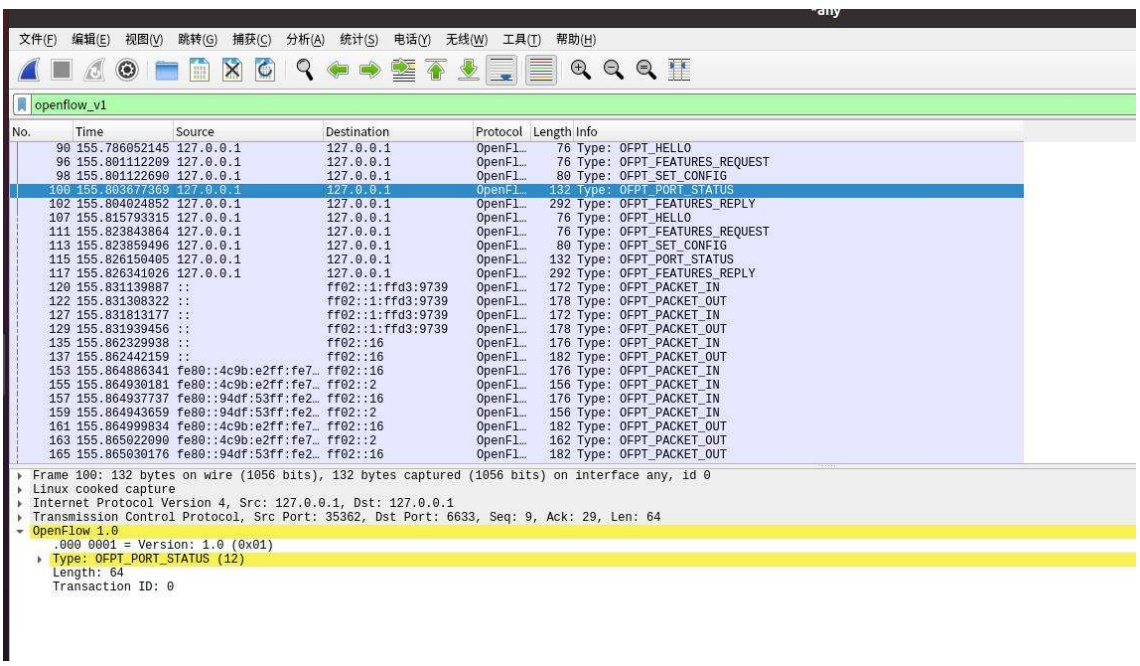
- Controller detects link failure
- Updates topology
- Logs "Link Removed"

---

7. Performance Evaluation


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
openflow_v1
No. Time Source Destination Protocol Length Info
42 12.694055640 127.0.0.1 127.0.0.1 OpenFlow 76 Type: OFPT_HELLO
44 12.694065759 127.0.0.1 127.0.0.1 OpenFlow 76 Type: OFPT_FEATURES_REQUEST
46 12.694071129 127.0.0.1 127.0.0.1 OpenFlow 80 Type: OFPT_SET_CONFIG
48 12.698668365 127.0.0.1 127.0.0.1 OpenFlow 244 Type: OFPT_FEATURES_REPLY
53 12.724182901 127.0.0.1 127.0.0.1 OpenFlow 76 Type: OFPT_HELLO
58 12.727058260 127.0.0.1 127.0.0.1 OpenFlow 76 Type: OFPT_FEATURES_REQUEST
60 12.727082665 127.0.0.1 127.0.0.1 OpenFlow 80 Type: OFPT_SET_CONFIG
62 12.730713009 127.0.0.1 127.0.0.1 OpenFlow 132 Type: OFPT_PORT_STATUS
64 12.731034311 127.0.0.1 127.0.0.1 OpenFlow 244 Type: OFPT_FEATURES_REPLY
67 12.777593587 :: ff02::16 OpenFlow 176 Type: OFPT_PACKET_IN
69 12.777666193 :: ff02::16 OpenFlow 182 Type: OFPT_PACKET_OUT
93 12.885324789 :: ff02::1:ffc7:6129 OpenFlow 172 Type: OFPT_PACKET_IN
95 12.885389550 :: ff02::1:ffc7:6129 OpenFlow 178 Type: OFPT_PACKET_OUT
99 12.905310082 :: ff02::16 OpenFlow 176 Type: OFPT_PACKET_IN
101 12.905391936 :: ff02::16 OpenFlow 182 Type: OFPT_PACKET_OUT
125 13.001495401 :: ff02::1:ff60:b49b OpenFlow 172 Type: OFPT_PACKET_IN
127 13.001584668 :: ff02::1:ff60:b49b OpenFlow 178 Type: OFPT_PACKET_OUT
n_tables: 254
Pad: 000000
capabilities: 0x000000c7
....1 = Flow statistics: True
....1. = Table statistics: True
.... .1.. = Port statistics: True
.... 0... = Group statistics: False
....0. .... = Can reassemble IP fragments: False
.... .1.. .... = Queue statistics: True
....0 .... = Switch will block looping ports: False
actions: 0x00000fff
....1 = Output to switch port: True
....1. = Set the 802.1q VLAN id: True
.... .1.. = Set the 802.1q priority: True
.... 1... = Strip the 802.1q header: True
....1 .... = Ethernet source address: True
....1. .... = Ethernet destination address: True
.... .1.. .... = IP source address: True
.... 1... .... = IP destination address: True
....1 .... = IP ToS (DSCP field, 6 bits): True
....1. .... = TCP/UDP source port: True
.... .1.. .... = TCP/UDP destination port: True
.... 1... .... = Output to queue: True
Port data 1
Port number: 65534
HW Address: ba:b3:68:b6:ef:46 (ba:b3:68:b6:ef:46)
Port Name: s1
Config flags: 0x00000001
State flags: 0x00000001
Current features: 0x00000000
Advertised features: 0x00000000
Features supported: 0x00000000
Features advertised by peer: 0x00000000
Port data 2
Port number: 1
HW Address: 66:78:b7:b4:4a:9e (66:78:b7:b4:4a:9e)
0000 00 00 03 04 00 06 00 00 00 00 00 00 00 00 08 00





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Using **Wireshark**:
Filter:

openflow_v4

Observed:

- OFPT_HELLO

- OFPT_FEATURES_REQUEST / REPLY
- OFPT_PACKET_IN / PACKET_OUT
- OFPT_PORT_STATUS

Result:

- Fast detection of link failures
- Efficient packet forwarding

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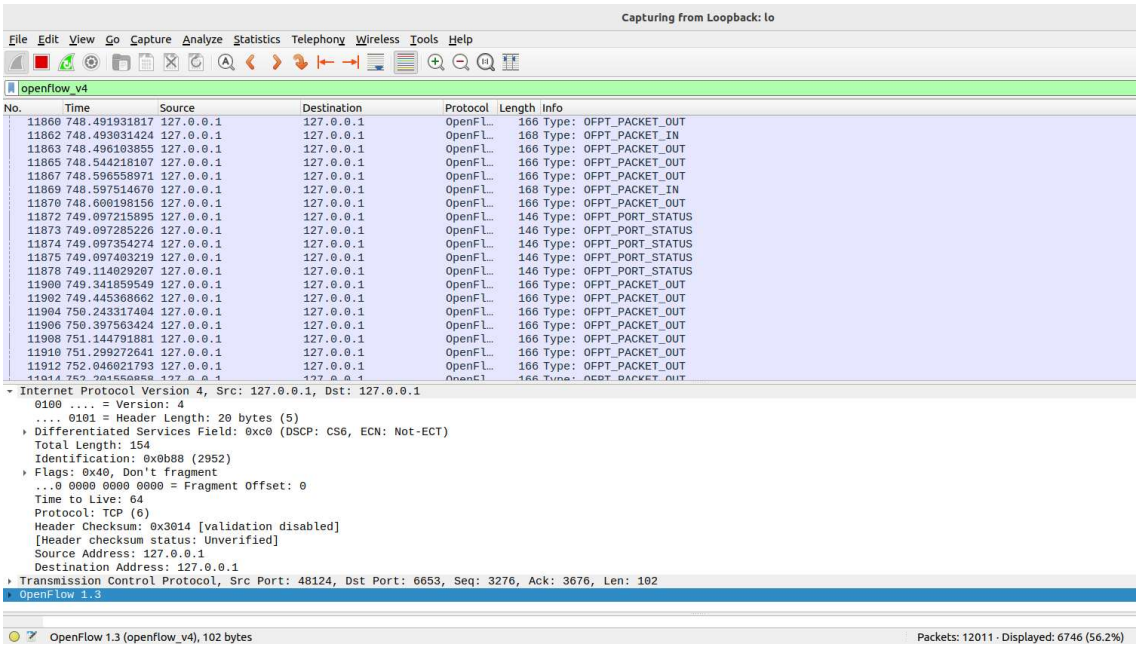
9. Screenshots of Demo (VERY IMPORTANT)




```
Link removed: Link: Port<dpid=1, port_no=2, DOWN> to Port<dpid=2, port_no=2, LIVE>
Current Switches: [1, 2]
Current Links: []
Link removed: Link: Port<dpid=2, port_no=2, LIVE> to Port<dpid=1, port_no=2, LIVE>
Current Switches: [1, 2]
Current Links: []
```

Capturing from Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
8749	550.193200217	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
8751	550.244625945	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
8753	550.245951845	127.0.0.1	127.0.0.1	OpenFl…	168	Type: OFPT_PACKET_IN
8754	550.248597032	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
8756	550.274113899	127.0.0.1	127.0.0.1	OpenFl…	150	Type: OFPT_PACKET_IN
8757	550.275479151	127.0.0.1	127.0.0.1	OpenFl…	148	Type: OFPT_PACKET_OUT
8758	550.276299849	127.0.0.1	127.0.0.1	OpenFl…	150	Type: OFPT_PACKET_IN
8759	550.277305073	127.0.0.1	127.0.0.1	OpenFl…	148	Type: OFPT_PACKET_OUT
8760	550.277575024	127.0.0.1	127.0.0.1	OpenFl…	150	Type: OFPT_PACKET_IN
8761	550.278711929	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
8762	550.279330392	127.0.0.1	127.0.0.1	OpenFl…	148	Type: OFPT_PACKET_OUT
8764	550.279592951	127.0.0.1	127.0.0.1	OpenFl…	150	Type: OFPT_PACKET_IN
8765	550.280613609	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
8766	550.280670303	127.0.0.1	127.0.0.1	OpenFl…	148	Type: OFPT_PACKET_OUT
8768	550.281066429	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
8769	550.282187994	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
8770	550.282248021	127.0.0.1	127.0.0.1	OpenFl…	204	Type: OFPT_PACKET_OUT
8772	550.282467469	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
8773	550.284384789	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
8774	550.284419049	127.0.0.1	127.0.0.1	OpenFl…	204	Type: OFPT_PACKET_OUT

```
▸ Frame 328: 168 bytes on wire (1344 bits), 168 bytes captured (1344 bits) on interface lo, id 0
▾ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Type: IPv4 (0x0800)
▾ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total Length: 154
    Identification: 0x0b88 (2952)
  ▸ Flags: 0x40, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x3014 [validation disabled]
    [Header checksum status: Unverified]
```

OpenFlow 1.3 (openflow_v4), 102 bytes Packets: 9723 · Displayed: 5457 (56.1%)

Capturing from Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
1369	80.502923705	127.0.0.1	127.0.0.1	OpenFl…	168	Type: OFPT_PACKET_IN
1370	80.506107944	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1372	81.247997946	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1374	81.299898302	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1376	81.301056415	127.0.0.1	127.0.0.1	OpenFl…	168	Type: OFPT_PACKET_IN
1377	81.304137142	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1379	81.352065567	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1381	81.404321117	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1383	81.405338269	127.0.0.1	127.0.0.1	OpenFl…	168	Type: OFPT_PACKET_IN
1384	81.408975379	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1386	82.056083629	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
1387	82.059725438	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
1389	82.149392496	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1391	82.200993509	127.0.0.1	127.0.0.1	OpenFl…	166	Type: OFPT_PACKET_OUT
1393	82.202085873	127.0.0.1	127.0.0.1	OpenFl…	168	Type: OFPT_PACKET_IN

```
▸ Frame 328: 168 bytes on wire (1344 bits), 168 bytes captured (1344 bits) on interface lo, id 0
▾ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Type: IPv4 (0x0800)
▾ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total Length: 154
    Identification: 0x0b88 (2952)
  ▸ Flags: 0x40, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x3014 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 48124, Dst Port: 6653, Seq: 3276, Ack: 3676, Len: 102
▸ OpenFlow 1.3
```

OpenFlow 1.3 (openflow_v4), 102 bytes Packets: 5773 · Displayed: 3194 (55.3%)

Capturing from Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
43	24.416166084	127.0.0.1	127.0.0.1	OpenFl…	98	Type: OFPT_FEATURES_REPLY
55	24.444141031	127.0.0.1	127.0.0.1	OpenFl…	98	Type: OFPT_FEATURES_REPLY
41	24.414625460	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_FEATURES_REQUEST
48	24.431016277	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_FEATURES_REQUEST
51	24.431299825	127.0.0.1	127.0.0.1	OpenFl…	146	Type: OFPT_FLOW_MOD
56	24.445627223	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
60	24.447434026	127.0.0.1	127.0.0.1	OpenFl…	146	Type: OFPT_FLOW_MOD
63	24.454785643	127.0.0.1	127.0.0.1	OpenFl…	162	Type: OFPT_FLOW_MOD
32	24.398565180	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_HELLO
37	24.413711553	127.0.0.1	127.0.0.1	OpenFl…	82	Type: OFPT_HELLO
39	24.413862832	127.0.0.1	127.0.0.1	OpenFl…	82	Type: OFPT_HELLO
46	24.430941155	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_HELLO
54	24.443651248	127.0.0.1	127.0.0.1	OpenFl…	274	Type: OFPT_MULTIPART_REPLY, OFPMP_PORT_DESC
62	24.453003162	127.0.0.1	127.0.0.1	OpenFl…	274	Type: OFPT_MULTIPART_REPLY, OFPMP_PORT_DESC
50	24.431283271	127.0.0.1	127.0.0.1	OpenFl…	82	Type: OFPT_MULTIPART_REQUEST, OFPMP_PORT_DESC

```
▸ Frame 328: 168 bytes on wire (1344 bits), 168 bytes captured (1344 bits) on interface lo, id 0
▾ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
  ▸ Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Type: IPv4 (0x0800)
▾ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total Length: 154
    Identification: 0x0b88 (2952)
  ▸ Flags: 0x40, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x3014 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 48124, Dst Port: 6653, Seq: 3276, Ack: 3676, Len: 102
▸ OpenFlow 1.3
```

OpenFlow 1.3 (openflow_v4), 102 bytes Packets: 2772 · Displayed: 1478 (53.3%)

```
Switch entered: 1
Current Switches: [1, 2]
Current Links: [(1, 2)]
Switch entered: 2
Current Switches: [1, 2]
Current Links: [(1, 2), (2, 1)]
Link added: Link: Port<dpid=1, port_no=2, LIVE> to Port<dpid=2, port_no=2, LIVE>
Current Switches: [1, 2]
Current Links: [(1, 2), (2, 1)]
Link added: Link: Port<dpid=2, port_no=2, LIVE> to Port<dpid=1, port_no=2, LIVE>
Current Switches: [1, 2]
Current Links: [(1, 2), (2, 1)]
```

```
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1 s2
*** Adding links:
(h1, s1) (h2, s2) (s2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
```

```
loading app topo_detect.py
loading app ryu.topology.switches
loading app ryu.controller.ofp_handler
instantiating app topo_detect.py of TopologyDetector
instantiating app ryu.topology.switches of Switches
instantiating app ryu.controller.ofp_handler of OFPHandler
```

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You must include:

1. Controller running
2. Mininet running
3. link s1 s2 down
4. Output showing "Link Removed"

---

10. GitHub Repository Link

https://github.com/saifulla821762-dev/AM807-CN-SDN-/blob/main/README%20(2).md

---

11. Conclusion

This project successfully demonstrates:

- Dynamic topology detection using SDN
- Real-time link failure detection
- Efficient packet forwarding using OpenFlow
- Practical understanding of controller-switch communication

Wireshark analysis confirms correct OpenFlow message exchange and validates system behavio